

Experimental study on thermal performance of a single-phase immersion cooling unit for high-density computing power data center

Xinshan Sun, Zhan Liu^{*}, Shenrui Ji, Kaifeng Yuan

State Key Laboratory of Intelligent Construction and Healthy Operation and Maintenance of Deep Underground Engineering, School of Mechanics and Civil Engineering, China University of Mining and Technology, Xuzhou 221116, China

ARTICLE INFO

Keywords:

Single-phase immersion cooling
High-density computing power
Data center
Thermal performance
Power usage effectiveness

ABSTRACT

Traditional air-cooling systems cannot meet special requirements of high-density computing power data centers (DCs). The single-phase immersion cooling (SPIC) is considered as one of the best ways to effectively cool high-density computing power DC cabinets. In this study, a SPIC unit was experimentally investigated with dielectric fluid Noah 3000D under a designed heat load of 20 kW. The impact of the cooling water flow rate q_{v2} on the thermal characteristics, flow resistance and power consumption of the SPIC unit were investigated. The results show that the temperatures of both the coolant and cooling water decrease with the increase of q_{v2} , but the temperature difference of coolant increases with the increase of q_{v2} . An increase in coolant temperature difference means an increase in temperature non-uniformity. The increase in q_{v2} also leads to an increase in system flow resistance. In this experiment, the pressure loss of the heat exchanger and pipelines on the coolant side, account for about 56 % and 44 %, respectively. The power usage effectiveness of the cooling system in DC (cPUE) and coefficient of performance (COP) of the SPIC unit vary in the range of 1.08–1.09 and 5.69–7.21, respectively. Moreover, it is found that simply increasing q_{v2} cannot significantly improve the heat exchange of SPIC units, but will increase system's energy consumption to a certain extent. The present study mainly focuses on the laboratory experimental test on the SPIC unit, and the related conclusions are significant to the design and operation of SPIC units in high-density computing power DCs.

1. Introduction

Under the advocacy and promotion of global digitization, digital technologies including big data, 5G, cloud computation, the internet of Things, and artificial intelligence are widely applied in communication, manufacturing, finance, transportation, healthcare, education and other fields. While new development momentum is injected into various industries, higher requirements are put forward for data storage, processing, and transmission capabilities. The number of data stored and manipulated by data centers (DCs) around the world shows a tendency of exponential growth. However, as energy-intensive enterprises, the high energy consumption of DCs becomes a difficulty to be addressed. It is reported that the energy consumption of DCs accounts for 0.9 % of the global total energy consumption in 2015, which will reach 4.5 % in 2025 and 8 % in 2030, respectively (Masanet et al., 2020). The energy consumption of cooling and power systems account for the majority in the total operation energy consumption of DCs. Almost 52 % of the power consumption comes from information technology (IT) devices, 38 % of it

comes from cooling systems and 10 % of it comes from other equipment (Salim and Tozer, 2010). It was evaluated that the power consumption of DCs in China was more than 400 billion kWh in 2023, which may account for 4 % of the nationwide total power consumption (Liu et al., 2021). The power usage effectiveness (PUE) is usually adopted to reflect the energy-saving degree of DC indirectly. At present, the PUE value in the most Chinese traditional air-cooling DCs ranges from 1.5 to 2.0 (Li et al., 2023a), which is far from the requirements of the industry and government. Therefore, there is great development potential in energy saving of DCs.

During the operation of data cabinets, the IT equipment dissipates huge heat to the ambient. The IT equipment cannot function properly if the temperature of the IT equipment is too high. When the IT equipment in important fields such as finance, goes down, it may cause enormous economic losses. Therefore, high-quality thermal management must be set on the operation and maintenance of DCs. The widely used heat dissipation method currently is air-cooling, which is roughly achieved by conveying cold air near IT equipment and exhausting hot air to realize heat removal. Due to advantages of strong universality, low

^{*} Corresponding author.

E-mail address: liuzhankd@cumt.edu.cn (Z. Liu).

<https://doi.org/10.1016/j.ijheatfluidflow.2024.109735>

Received 16 May 2024; Received in revised form 6 November 2024; Accepted 22 December 2024

Available online 25 December 2024

0142-727X/© 2024 Elsevier Inc. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

Nomenclature			
A	Heat exchange area, m^2	v_1	Inlet velocity of coolant in coolant circulation pump, m/s
c_1	Specific heat capacity of coolant, $\text{kJ}/(\text{kg}\cdot\text{K})$	v_2	Outlet velocity of coolant in coolant circulation pump, m/s
c_2	Specific heat capacity of cooling water, $\text{kJ}/(\text{kg}\cdot\text{K})$	W_R	Power consumption for system refrigeration, kW
d	Diameter of pipeline, m	Z_1	Inlet elevation of coolant in coolant circulation pump, m
g	Gravitational acceleration, m/s^2	Z_2	Outlet elevation of coolant in coolant circulation pump, m
k	Heat transfer coefficient, $\text{W}/(\text{m}^2\cdot\text{K})$	<i>Greek symbols</i>	
k_T	Temperature change rate	α_1	Thermal diffusivity coefficient of coolant, m^2/s
$k_{\Delta p}$	Pressure drop change rate	Δp	Pressure difference, kPa
l	Length of the pipeline, m	Δp_1	Pressure difference between inlet and outlet of coolant circulation pump, kPa
L	Instrument range	Δp_2	Pressure difference between inlet and outlet of coolant in heat exchanger, kPa
M	Accuracy class	Δp_3	Pressure difference between inlet and outlet of cooling water in heat exchanger, kPa
m_1	Mass flow rate of coolant, kg/s	ΔT	Logarithmic average temperature difference, $^\circ\text{C}$
m_2	Mass flow rate of cooling water, kg/s	ΔT_c	Temperature difference between inlet and outlet of coolant, $^\circ\text{C}$
p_1	Inlet pressure of coolant in coolant circulation pump, kPa	ΔT_w	Temperature difference between inlet and outlet of cooling water, $^\circ\text{C}$
p_2	Outlet pressure of coolant in coolant circulation pump, kPa	Δx	Absolute error
p_3	Inlet pressure of coolant in heat exchanger, kPa	Δx_T	Absolute error of temperature sensor, $^\circ\text{C}$
p_4	Outlet pressure of coolant in heat exchanger, kPa	η	Efficiency of coolant circulation pump
p_5	Outlet pressure of cooling water in heat exchanger, kPa	λ	Frictional resistance coefficient
p_6	Inlet pressure of cooling water in heat exchanger, kPa	λ_1	Thermal conductivity of coolant, $\text{W}/(\text{m}\cdot\text{K})$
P	Energy provided by pump, kPa	ξ	Local resistance coefficient
P_e	Effective power of coolant circulation pump, kW	ρ	Density, kg/m^3
P_f	Frictional resistance loss, kPa	ρ_1	Inlet density of coolant in coolant circulation pump, kg/m^3
P_h	Rated heating power, kW	ρ_2	Outlet density of coolant in coolant circulation pump, kg/m^3
P_{IT}	Power consumption of IT equipment, kW	ρ_3	Density of coolant, kg/m^3
p_m	Local resistance loss, kPa	ρ_4	Density of cooling water, kg/m^3
P_R	Refrigeration capacity of system, kW	<i>Abbreviation</i>	
P_{total}	Total power consumption of system, kW	COP	Coefficient of Performance
Q_1	Heat exchange capacity on the coolant side, kW	cPUE	Power Usage Effectiveness of the cooling system in data center
Q_2	Heat exchange capacity on the cooling water side, kW	DC	Data Center
q_{v1}	Volume flow rate of coolant, m^3/s	IT	Information Technology
q_{v2}	Volume flow rate of cooling water, m^3/s	PUE	Power Usage Effectiveness
T	Temperature, $^\circ\text{C}$	SPIC	Single-Phase Immersion Cooling
$T_{c.in}$	Inlet temperature of coolant, $^\circ\text{C}$	TPIC	Two-Phase Immersion Cooling
$T_{c.out}$	Outlet temperature of coolant, $^\circ\text{C}$		
$T_{w.out}$	Outlet temperature of cooling water, $^\circ\text{C}$		
$T_{w.in}$	Inlet temperature of cooling water, $^\circ\text{C}$		
t	Time, s		
U_{N1}	Uncertainty of heat exchange capacity, kW		
U_{N2}	Uncertainty of heat transfer coefficient, $\text{W}/(\text{m}^2\cdot\text{K})$		
U_{N3}	Uncertainty of thermal diffusivity coefficient, m^2/s		
v	Velocity, m/s		

initial investment and low maintenance costs, the air-cooling method is therefore widely adopted in DCs (Brill, 2007; Landauer, 1961; Zhou et al., 2019; Gładyszewska-Fiedoruk and Teleszewski, 2023). However, the lifespan and reliability of IT equipment are easily influenced by the operation parameters, such as air temperature, humidity, and cleanliness (Khattar, 2014). Moreover, due to the limitations of air-cooling, such as the mixing of cold and hot air flows, the system has low cooling efficiency and high energy consumption. Many researchers proposed corresponding measures for improvement of IT cooling, including adding heat pipes in radiators (Mohammed et al., 2012), using evaporative coolers (Duan et al., 2012), constructing containment systems to strengthen airflow management (Isazadeh et al., 2023) and others. However, the air-cooling units will reach the limit of development, and it is difficult to meet the heat dissipation requirement of high power density IT devices, especially microprocessors (Brill, 2007).

Compared with air-cooling systems, the liquid-cooling systems are popular in the field of DCs for high heat transfer efficiency, great heat dissipation uniformity, low noise and high energy efficiency. The liquid-cooling can be divided into direct cooling and indirect cooling. Indirect

liquid-cooling is that the cooled device does not directly contact with coolants, typified by cold plate cooling, and the related systems are intricate and complex (Lee et al., 2015; Singh et al., 2009). Cold plate cooling is commonly combined with air-cooling to improve the total heat dissipation efficiency of the servers by strengthening heat dissipation of high-power electronic chips in local regions. Direct liquid-cooling is that the servers are directly immersed in coolants, whose typical representative is immersion-cooling. Compared with air-cooling systems, immersion-cooling operates without air-conditioning system, and can satisfy the demand of cooling servers with power more than 5 kW (Kanbur et al., 2021). The immersion cooling systems are classified as single-phase and two-phase immersion cooling. The single-phase immersion cooling (SPIC) means that the coolant only transfers heat in the form of liquid phase, while two-phase immersion cooling (TPIC) exchanges heat by latent heat of low-temperature boiling fluids. Kanbur et al. (2021) compared and analyzed thermodynamic performance of a SPIC and TPIC systems, and found that the COP of the TPIC system is 72.2–79.3 % higher than that of the SPIC system. The TPIC systems have greater heat exchange capability than SPIC systems, and also have

higher investment, worse safety and lower maintainability. However, the flow stability of TPIC is relatively poor, which usually leads to thermo-flow behaviors such as super-heating and burnout of the heated surfaces (Kadam and Kumar, 2014). Therefore, the SPIC systems have wide range of utilization in large-scale and commercial cooling fields. Luiten (2021) simulated the thermal performance of SPIC servers by using hydrocarbons as coolant. After detailed optimization, the volumetric cooling performance of a SPIC server was improved by nearly three times. Shah et al. (2019) immersed server components such as circuit boards and cables into mineral oil and synthetic dielectric fluids to study the compatibility between servers and coolant. It was found that the electrical characteristics of passive components are completely within an acceptable range at an ambient temperature of 45 °C. Pambudi et al. (2022) used mineral oil and virgin coconut oil as coolant to research the performance of SPIC systems. They found that the CPU's temperature of the SPIC system is 13 % lower than that of traditional air-cooling system. The flow rate has a significant impact on temperature, and virgin coconut oil has a better cooling effect than mineral oil. After soaking CPU in coolant for 5 months, the CPU could still function normally, indicating strong compatibility between the electric material and coolant. Gajjar (2023) developed a new SPIC system consisting of a multi-staged Tesla valve filled with dielectric liquid, which was driven only by thermal convection without the assistance of a pump. They selected mineral oil and FC-3283 as coolant and conducted CFD simulations to validate the feasibility of the SPIC system. The results showed that the temperature of CPU is within an appropriate range in both cases. Compared to mineral oil, FC-3283 has a better cooling effect. Taddeo et al. (2023) established an energy model of a SPIC system, and explored the model response to dynamic and static workloads. Huang et al. (2023) conducted a simulation experiment on buoyancy-driven and pump-driven SPIC tanks, and verified that the average temperature of coolants and PUE of pump-driven SPIC systems are 55.5 % and 11.6 % lower than that of buoyancy-driven systems. Li et al. (2023b) improved the performance of SPIC servers by optimizing the parameters like fin height, fin spacing, thermal conductivity, height ratio of baffle, outlet area, inlet flow rate and inlet temperature. Cheng et al. (2020) tested the simulation model of a forced-circulation SPIC system with heat sinks using different materials, and concluded that the materials of heat sinks have slight influences on the cooling effect of the SPIC.

Based on above introductions, it is easy to find that the SPIC system has already been tested numerically and experimentally, and most of the research is conducted at the micro-scale level or server level (Luiten, 2021; Shah et al., 2019; Pambudi et al., 2022; Gajjar, 2023; Huang et al., 2023; Li et al., 2023b). However, in practical engineering applications, multiple servers are often combined into one cabinet for saving space. Although different investigations have been conducted on the SPIC system, there is still no comprehensive experimental investigation on the cabinet level SPIC system from the perspective of practical engineering application, which brings a remarkable research gap from the academic perspective.

To make up the research gap mentioned above, the present work conducted detailed experimental investigation on a SPIC unit applied to high-density computing power DCs. The actual equipment utilized in the construction of DCs is adopted to construct the SPIC unit. To simulate high heat release capacity from actual servers, the electric heating device is adopted to supply a rated heating power of 20 kW. Due to direct contact between the coolant and the equipment during immersion-cooling, the thermal characteristics of the coolant directly affect the cooling effect (Jithin and Rajesh, 2022). The dielectric fluid Noah 3000D is selected as the circulation working fluid. The thermal behavior of the established SPIC unit is experimentally researched under a designed heat load of 20 kW and different cooling water flow rates. Some valuable findings are obtained. This work can supply some technique references for the design and operation of SPIC units in high-density computing power DCs.

2. Experimental rig and test procedure

The schematic diagram and photo of the present SPIC experimental rig (Liu et al., 2024) are shown in Fig. 1 and Fig. 2. It is easy to see the experimental system mainly consists of a primary side cooling subsystem, a secondary side cooling subsystem, and a monitoring subsystem. The experiment test was conducted in a constant temperature and humidity chamber, which could supply temperature range of 21–23 °C and relative humidity range of 40–60 %. The primary side cooling subsystem is located in an outdoor environment, utilizing direct heat exchange between external air and cooling water to generate low-temperature cooling water, which dissipates the heat generated by electronic devices to the ambient. The secondary side cooling subsystem and monitoring subsystem are placed in the liquid cooling cabinet.

2.1. Experimental rig

The primary side cooling subsystem consists of an open cooling tower, a cooling water circulation pump, a heat exchanger, two turbine flow sensors and connection pipelines. In heat exchanger, the heat from the high-temperature coolant is transferred to the low-temperature cooling water, which is provided by releasing the heat to the ambient through an open cooling tower. The secondary side cooling subsystem consists of a tank, a coolant circulation pump, a heat exchanger, two turbine flow sensors and connection pipelines. Electronic devices are directly immersed in a tank filled with dielectric fluid. The operation of electronic devices generates and releases heat, and the dielectric fluid brings the heat release to the heat exchanger with a temperature rise. The monitoring subsystem consists of temperature sensors, pressure sensors, turbine flow meters, liquid level sensors, and a multifunctional data acquisition instrument, used for real-time monitoring and recording of experimental data.

The test tank is made of stainless steel plates. The size of the test tank is 580 × 620 × 1050 mm (W × L × H). The Noah 3000D is selected as the dielectric fluid, which has good thermodynamic performance and insulation characteristics. The physical parameters of the selected Noah 3000D is listed in Table 1. To simulate high heat release rates from servers, the electric heating device is adopted to supply a rated heating power of 20 kW. A horizontal stainless steel multi-stage shielded pump is utilized as the coolant circulation pump with a rated flow rate of 4 m³/h, and is used to drive the circulation of coolant in the secondary side cooling subsystem. A copper brazed plate heat exchanger is chosen as the heat exchange device with a heat exchange area of about 1.49 m². In heat exchanger, the heat from the high-temperature coolant of the secondary side cooling subsystem is transferred to the low-temperature cooling water of the primary side cooling subsystem. A vertical centrifugal pump with a rated flow rate of 25 m³/h and a rated head of 20 m, is selected as the cooling water circulation pump in the primary side cooling subsystem. An open circular counter flow industrial type cooling tower is used as heat release device, which uses an axial flow fan to induce air to contact and exchange heat with the cooling water sprayed down, cooling the cooling water.

2.2. Test monitors arrangement

The PT100 thermistors are utilized as temperature sensors in monitoring system, which are inserted into connection pipelines. The test range of the selected Platinum resistance sensor is –50–200 °C and the test accuracy is A-level. The universal pressure transmitters with a pressure measurement range of –0.1–0.6 MPa and an accuracy of 0.2 %, are adopted in experiment. Both the temperature and pressure sensors are arranged at the inlet and outlet of the heat exchanger in the primary and secondary side cooling subsystems. Two pressure sensors are installed at the inlet and outlet of the coolant circulation pump. The measurement range of the turbine flow meters is 2–20 m³/h, with a test accuracy of 0.5 %. The turbine flow sensors are installed in both the

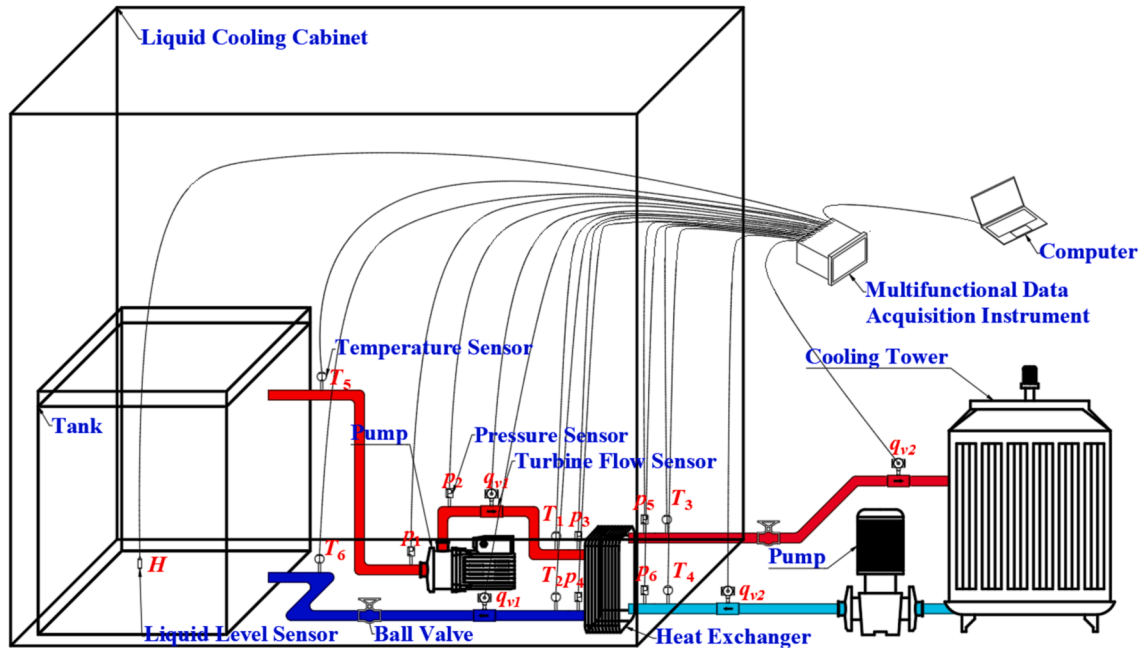


Fig. 1. Schematic diagram of the experimental system (Liu et al., 2024).

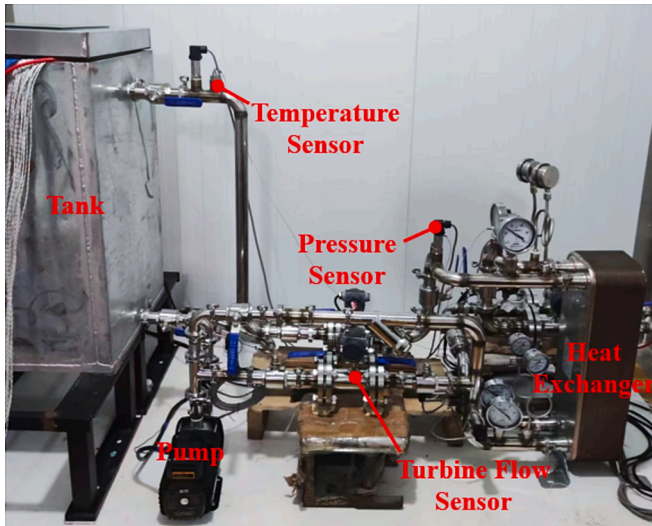


Fig. 2. Photo of the test system (Liu et al., 2024).

Table 1

Physical parameters of Noah 3000D at atmospheric pressure and 25 °C.

Physical parameters	Values
Boiling point (°C)	112
Liquid density (kg/m ³)	1815
Thermal conductivity (W/(m•K))	0.0623
Kinematic viscosity(m ² /s)	1.353×10^{-6}
Special heat (J/(kg•K))	1177

primary and secondary side cooling loops. An input type liquid level sensor is used to test the variation of liquid height in the test tank, with a range of 0–3 m and an accuracy of 0.5 %. All test data is collected by the multifunctional data acquisition instrument. The multifunctional data acquisition instrument has 36 acquisition channels and a sampling frequency of 1 Hz.

2.3. Test procedure

In the present experiment test, the total heat load is set as 20 kW. The coolant flow rate of all three cases maintains at 6.2 m³/h, and the thermal performance of the SPIC system is evaluated by changing the circulation flow rate of cooling water. The circulation flow rates of cooling water in cases No.1–3 are 4.4 m³/h, 5.4 m³/h and 6.4 m³/h, respectively.

The experimental testing procedure mainly include two stages, namely pre-treatment and formal testing. The following work needs to be done during the pre-treatment stage of the experiment.

- (1) The airtightness test is conducted to ensure the tightness of the experimental device.
- (2) All experimental equipment is inspected and calibrated.
- (3) Repetitive experiments need to be conducted. The reproducibility of the experiment should be verified before formal tests.

When everything is normal, the formal tests can be carried out. The formal testing steps are given as follows.

- (1) The circulating flow rates of coolant and cooling water are adjusted to the set values through the control valve.
- (2) After the experimental device operates stably, the electric heating device is activated and the rated heating power is set to be 20 kW.
- (3) After the experimental device operates stably for a period of time, the electric heating device is turned off and the experimental device is allowed to operate for another period of time.
- (4) The test is completed and the data is saved in the multifunctional data acquisition instrument.
- (5) Finally, the pipeline valve is closed and the power supply to the equipment is cut off.

3. Data processing and error analysis

3.1. Data processing

All experimental data is collected by a multifunctional data acquisition instrument, and the monitored results are imported into Excel for

data processing using MATLAB software. During the experiment, abnormal test data may be caused by malfunctions in the testing device, unexpected changes in measurement conditions, and strong external interference. In processing data, the Laida criterion is used to eliminate the abnormal data. In addition, the fluctuations in temperature and humidity inside the laboratory, and the fluctuations in the voltage of the power supply used for monitoring instruments, also occur in the experimental test. The random errors are usually caused by these small disturbances. After verification, the random error of the measured values in this experiment approximately follows a normal distribution, and its arithmetic mean can be used as the best estimate of the true value. In this study, the arithmetic mean of 60 data points within a minute is treated as the true value of the measured value within each minute.

3.2. Error analysis

The experiment instrument can cause some certain errors, and the formula for calculating the total error of each instrument is expressed by equations (1) and (2).

$$\Delta x_T = \pm(0.15 + 0.002 \times |T|) \quad (1)$$

where, Δx_T is the total error of the A accuracy class temperature sensor, and T is the measured value of the temperature sensor.

$$\Delta x = \pm L \times M\% \quad (2)$$

where, Δx is the total error of another instrument, L is the instrument range, and M is the accuracy class.

The detailed test range and accuracy of different test sensors are listed in Table 2.

When the experimental operation reaches stable state, the random error distribution of the measured values can be obtained by subtracting the average of the measured values from the measured values. The Random errors follow statistical distribution laws and can be estimated using statistical methods. The random error distributions of all experimental measurement values are tested, and the random error distributions of some experimental measurement values are shown in Fig. 3. It can be found that the random error of the measurement values roughly follows a normal distribution.

Here, the Laida criterion is used to eliminate the abnormal data. The principle of the Laida criterion is that when only random errors exist in some experiments, the random errors of the experimental data mostly follow a normal distribution or approximate normal distribution. By applying statistical principles, the standard deviation can be determined. According to statistical principles, the probability of the measured value falling within the range of plus or minus three times the standard deviation is 99.7 %. Deviations greater than three times the standard deviation are considered outliers and should be removed.

Direct measurement values have test errors, and indirect measurement values obtained through function operation from direct measurement values also have errors, which is the propagation of errors. The functional relationship, between the indirect measurement quantity N and the independent direct measurement quantities $x, y, z, \dots$, can be expressed as

$$N = f(x, y, z, \dots) \quad (3)$$

Table 2
Accuracy class and error range of measuring instruments.

Instrument name	Instrument range	Accuracy class	Error range
Temperature sensor	−50–200 °C	A	$\leq \pm 0.55$ °C
Pressure sensor	−100–600 kPa	0.2 %FS	$\leq \pm 1.4$ kPa
Turbine flow sensor	2–20 m ³ /h	0.5 %FS	$\leq \pm 0.09$ m ³ /h
Liquid level sensor	0–3 m	0.5 %FS	$\leq \pm 0.015$ m

The calculation formula for the relative uncertainty is determined by

$$E_N = \frac{U_N}{N} = \sqrt{\left(\frac{\partial \ln f}{\partial x}\right)^2 U_x^2 + \left(\frac{\partial \ln f}{\partial y}\right)^2 U_y^2 + \left(\frac{\partial \ln f}{\partial z}\right)^2 U_z^2 + \dots} \quad (4)$$

where, E_N is the relative uncertainty of the indirect measurement quantity N , U_N is the uncertainty of the indirect measurement quantity N , and U_x , U_y and U_z represent the uncertainty of the direct measurement quantities x, y and z , respectively.

4. Experimental results

Under different test settings, the variations of fluid temperature and temperature change rate are considered and introduced. The temperature change rate can reflect the severity of fluid temperature fluctuations and is determined by equation (5).

$$k_T = dT/dt \quad (5)$$

where, k_T is the temperature change rate, T is the temperature value, and t is the time.

Generally, temperature fluctuations can affect the electrical performance of electronic components, and changes in these parameters may affect the normal operation of electronic components and even lead to electrical failures. The corresponding fluid temperature variations of three cases during the whole test process are shown in Fig. 4(a), Fig. 5(a) and Fig. 6(a). It is clear to see during the initial stage of the experiment, the fluid temperature change rate fluctuates around 0. When the heat load is imposed and set as 20 kW, the temperature change rate first suddenly increases, then gradually decreases, and finally decreases to around 0. After a period of time, when the heat load is turned off, the temperature change rate undergoes suddenly decrease, then gradually increases, and finally tends to be around 0. For different cases, the change rate of coolant inlet temperature is higher than that of coolant outlet temperature. Therefore, when designing the secondary side cooling subsystem, the electronic components should be placed as close as possible to the coolant outlet position to reduce the impact of the heat exchange with the ambient.

Fig. 4(b), Fig. 5(b) and Fig. 6(b) show the temperature variations of different monitors throughout the entire experimental process. For different test cases, the temperature fluctuates slightly due to changes in ambient temperature during the initial stage of the experiment test. When a heat load of 20 kW is activated and imposed, the fluid temperature gradually increases and finally fluctuates around a certain value. At this point, the system reaches a stable state at a rated heat load of 20 kW. After a period of stable operation, the system is powered off, and the fluid temperature undergoes a gradual reduction. It is easy to find that the inlet temperature of the coolant is higher than the outlet temperature of the coolant, and the outlet temperature of the coolant is higher than the outlet temperature of the cooling water. The inlet temperature of the cooling water is the lowest. For different cases, the fluid temperature decreases with the increase of the circulation flow rate of cooling water.

Under different test settings, the variations of fluid pressure drop change and the pressure drop change rate are introduced. The pressure drop change rate is determined by equation (6).

$$k_{\Delta p} = d\Delta p/dt \quad (6)$$

where, $k_{\Delta p}$ is the pressure drop change rate, and Δp is the pressure drop value.

Fig. 7 displays the pressure drop (inlet pressure minus outlet pressure) change rate of coolant in coolant circulation pump (Δp_1) and heat exchanger (Δp_2) during the whole test process under three cases. The pressure drop change may cause impacts on electronic components and affect the normal operation of the SPIC system. It is easy to see during the initial stage of the experiment, the pressure drop change rate

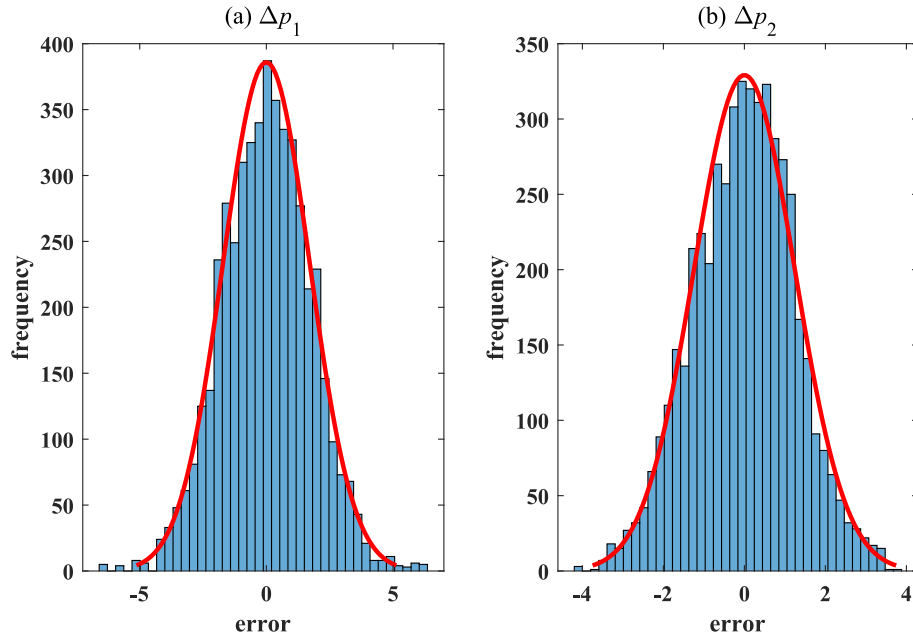


Fig. 3. Random error distributions.

fluctuates around 0. Once the heat load is activated, the pressure drop change rate first rapidly decreases to the minimum value, and then sequentially increases. After more than ten minutes, the pressure drop change rate returns to fluctuate around 0. When it comes into the cooling stage, the pressure drop change rate increases first, and then decreases to fluctuate around 0. Generally, the pressure drop change trend in cooling stage is opposite to that in heating stage. For different cases, the pressure drop change rate of coolant in coolant circulation pump is higher than that in heat exchanger. Therefore, when designing the secondary side cooling subsystem, the electronic components should be located as far away as possible from the coolant circulation pump to reduce the impact of pressure fluctuations on the electronic components.

Fig. 8 shows the pressure drop variations of coolant circulation pump and heat exchanger during the whole experimental process under three cases. Δp_3 represents the pressure drop of cooling water in heat exchanger. Throughout the whole experimental process, both Δp_1 and Δp_2 decrease at the beginning of heating stage, then stabilize at a certain value, and finally increase at the beginning of cooling stage. However, the variation of Δp_3 is not significant throughout the entire experimental process. When the system reaches a stable state in heating stage, with cooling water flow rate varying from 4.4 m³/h to 6.4 m³/h, Δp_1 slightly increases from 182 kPa to 185 kPa, Δp_2 increases from 101 kPa to 103 kPa, and Δp_3 significantly increases from 29 kPa to 56 kPa. Under different cases, as the circulation flow rate of cooling water increases, the pressure drop gradually increases. The reasons for pressure changes are explained as follows. The pressure drop is calculated as

$$p_f = \lambda \frac{l}{d} \frac{\rho v^2}{2} \quad (7)$$

$$p_m = \xi \frac{\rho v^2}{2} \quad (8)$$

where, p_f is the frictional resistance loss, p_m is the local resistance loss, λ is the frictional resistance coefficient, ξ is the local resistance coefficient, l is the length of the pipeline, d is the diameter of the pipeline, ρ is the density, and v is the velocity.

According to equations (7) and (8), under certain pipeline characteristics, the pressure loss is proportional to fluid density and the square of flow velocity. Due to the coolant flow rate of all three cases maintains at 6.2 m³/h, the changes in Δp_1 and Δp_2 can only be caused by changes

in coolant density. When the rated heat load is set as 20 kW, the fluid temperature increases and the fluid density decreases, which causes the reduction of the pressure loss. Therefore, selecting a liquid with a relatively large density variation with temperature as the coolant can reduce the flow resistance.

The energy conservation equation for the fluid column from section 1 of the pump inlet pressure measurement point to section 2 of the pump outlet pressure measurement point is expressed as

$$P = \left(p_2 + \frac{\rho_2 v_2^2}{2} + \rho_2 g Z_2 \right) - \left(p_1 + \frac{\rho_1 v_1^2}{2} + \rho_1 g Z_1 \right) \quad (9)$$

where, P is the energy provided by the pump, p_2 and p_1 represent the pressure values of sections 2 and 1, ρ_2 and ρ_1 represent the density values of sections 2 and 1, v_2 and v_1 represent the velocity values of sections 2 and 1, Z_2 and Z_1 represent the elevations of sections 2 and 1, respectively, and g is the gravitational acceleration.

When the density and flow velocity of the fluid in two sections are almost the same, the difference in dynamic pressure and potential pressure between the two sections is basically zero. The pressure difference between the inlet and outlet of the pump is approximately equal to the pressure head of the pump. According to the conservation of energy, the pressure drop at the inlet and outlet of the pump is approximately equal to the total pressure drop of the system. Based on the experimental results, it can be calculated that on the coolant side, the pressure loss of the heat exchanger accounts for about 56 %, and the pressure loss of pipelines and other equipment accounts for about 44 %. Therefore, a well-designed coolant pipeline is benefit to reduce energy loss.

5. Analysis and discussion

5.1. Thermal characteristics

To investigate the thermal characteristics of the present SPIC unit, the specific indirect variables such as logarithmic average temperature difference, need to be calculated using first-hand data obtained from experiment tests.

$$\Delta T = \frac{(T_{c.in} - T_{w.out}) - (T_{c.out} - T_{w.in})}{\ln \frac{T_{c.in} - T_{w.out}}{T_{c.out} - T_{w.in}}} \quad (10)$$

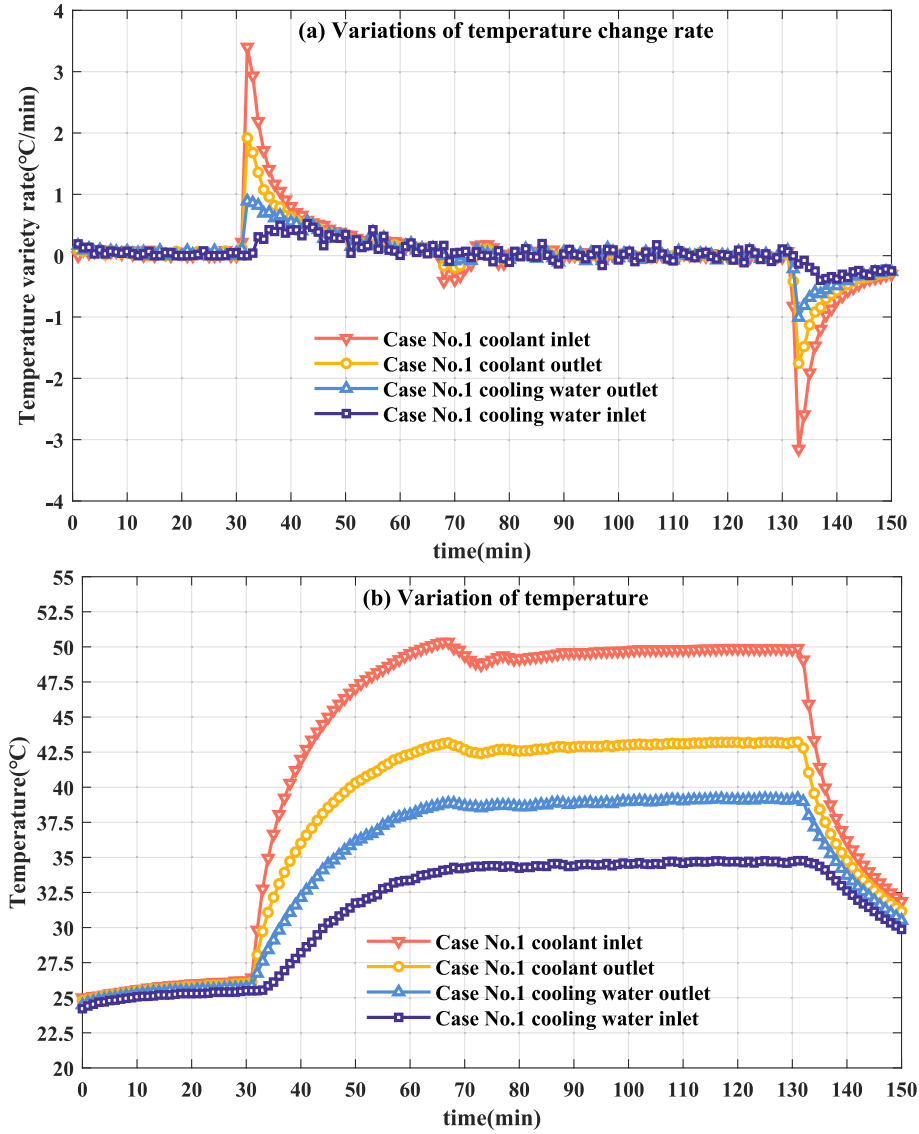


Fig. 4. Fluid temperature variations of Case No.1.

where, ΔT is the logarithmic average temperature difference, $T_{c.in}$ is the inlet temperature of the coolant, $T_{c.out}$ is the outlet temperature of the coolant, $T_{w.out}$ is the outlet temperature of the cooling water, and $T_{w.in}$ is the inlet temperature of the cooling water.

When the stable situation is obtained, different monitored temperature values are collected and listed in Table 3. The indirect temperature values such as logarithmic average temperature difference are calculated and listed in Table 4.

The temperature values are plotted in Fig. 9 to provide a more intuitive analysis of the impact of the cooling water flow rate on fluid temperature variations.

Based on Table 3, Table 4 and Fig. 9, it can be seen that in stable state, as the cooling water flow rate q_{v2} increases, $T_{c.in}$, $T_{c.out}$, $T_{w.out}$ and $T_{w.in}$ all decrease. But the four temperature drops are not consistent and the change in $T_{w.out}$ is most significant. Comparing case No.3 with case No.1, when q_{v2} increases by 45.45 %, $T_{w.out}$ decreases by 10.74 %. It is easy to understand that ΔT_w decreases as q_{v2} increases. However, ΔT_c increases with the increase of q_{v2} due to changes in physical parameters of coolant. If ΔT_c is too large, it may cause uneven temperature distributions in the tank. Therefore, in consideration of the variations of coolant physical parameters with temperature, it is necessary to select a suitable coolant.

The heat exchange capacity, heat transfer coefficient, and thermal diffusivity coefficient are calculated and analyzed as well. The heat exchange capacity Q_1 on the coolant side is calculated as

$$Q_1 = c_1 m_1 \Delta T_c = c_1 q_{v1} \rho_3 \Delta T_c \quad (11)$$

where, c_1 is the specific heat capacity of coolant, with a value of 1.177 kJ/(kg·°C), m_1 is the mass flow rate of coolant, kg/s, q_{v1} is the volume flow rate of coolant, m³/s, and ρ_3 is the density of coolant, with a value of 1815 kg/m³.

The heat exchange capacity Q_2 on the cooling water side is calculated as

$$Q_2 = c_2 m_2 \Delta T_w = c_2 q_{v2} \rho_4 \Delta T_w \quad (12)$$

where, c_2 is the specific heat capacity of cooling water, with a value of 4.1813 kJ/(kg·°C), m_2 is the mass flow rate of cooling water, kg/s, q_{v2} is the volume flow rate of cooling water, m³/s, and ρ_4 is the density of cooling water, with a value of 997.05 kg/m³.

The total heat transfer coefficient k of the heat exchanger is determined by

$$k = \frac{Q_1 + Q_2}{2A\Delta T} \quad (13)$$

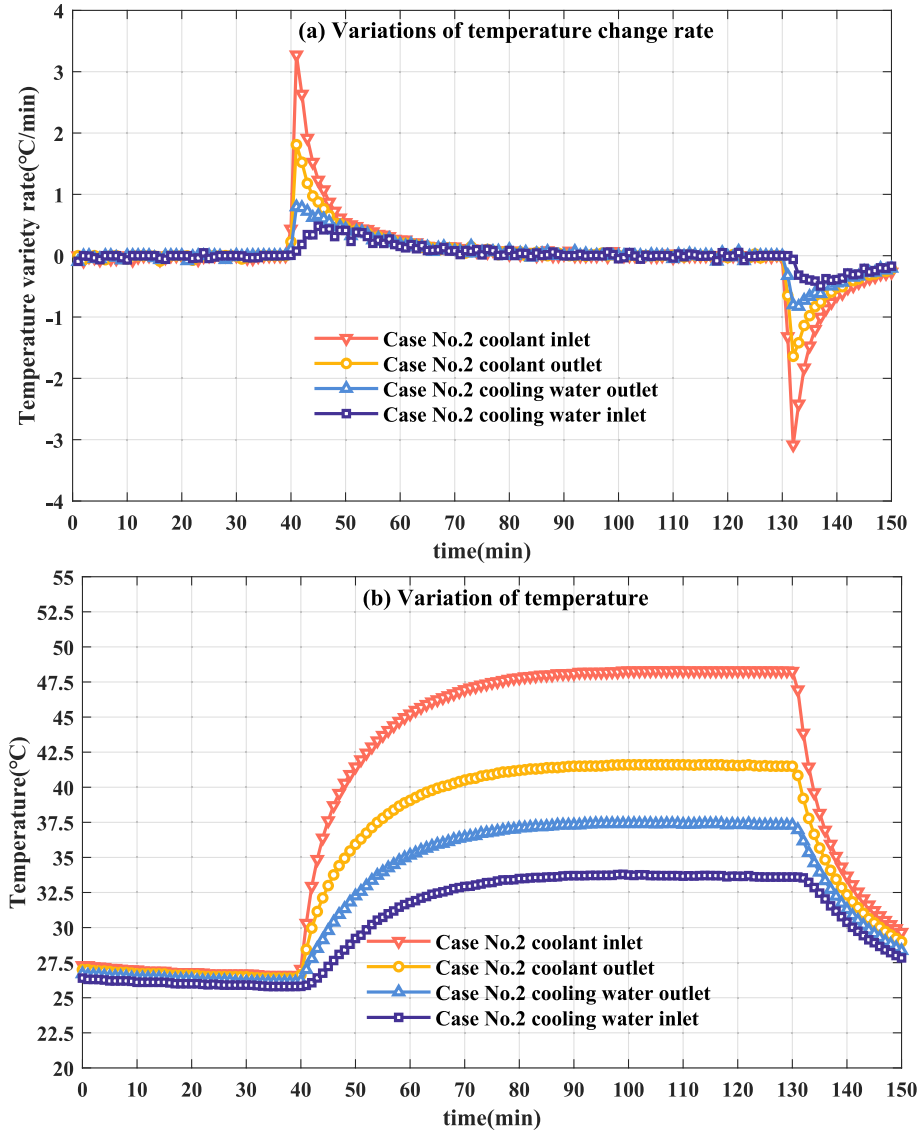


Fig. 5. Fluid temperature variations of Case No.2.

where, A is the heat exchange area, taken as 1.49 m^2 (the heat exchange area of the heat exchanger).

The thermal diffusivity coefficient α_1 of the coolant can be calculated by

$$\alpha_1 = \frac{\lambda_1}{\rho_3 c_1} = \frac{2\lambda_1}{Q_1 + Q_2} q_{v1} \Delta T_c \quad (14)$$

where, α_1 is the thermal diffusivity coefficient of the coolant, m^2/s , and λ_1 is the thermal conductivity of the coolant, taken as $0.0623 \text{ W}/(\text{m}\cdot\text{K})$.

Some thermal parameters such as heat exchange capacity, can be found in Table 5. It is easy to see the deviation between Q_1 and Q_2 does not exceed 10 %. The main reason for the deviation is because the temperature changes cause changes in fluid properties. It can be seen that the heat exchange capacity of the three cases is higher than the total heat load. The possible reasons are explained as follows. Firstly, the coolant circulation pump adopts shielded motor wet rotor technology, which has advantages such as low noise and no leakage. However, the cooling of the motor is borne by the coolant. Secondly, due to voltage fluctuations, the heating device generates more heat than the rated power. There is evidence that the SPIC unit may be able to meet greater heat dissipation requirements.

The uncertainties of heat exchange capacity, heat transfer coefficient, and thermal diffusivity coefficient are listed in Table 6.

Fig. 10 shows variations of the heat exchange capacity and heat transfer coefficient under different cooling water flow rates. It is easy to see as the cooling water flow rate increases, both the heat exchange capacity and heat transfer coefficient experience slightly increase. The heat exchange capacity is around 24 kW, the heat transfer coefficient is between $1648\text{--}1751 \text{ W}/(\text{m}^2\cdot\text{K})$, and the thermal diffusivity coefficient is about $3 \times 10^{-8} \text{ m}^2/\text{s}$. The thermal diffusivity coefficient of water is approximately $14 \times 10^{-8} \text{ m}^2/\text{s}$, which is 4 times higher than the thermal diffusivity coefficient of the dielectric fluid. The thermal diffusivity coefficient represents the ability of an object to achieve uniform temperature during heating or cooling. To prevent damage to electronic components caused by uneven temperature, liquids with higher thermal diffusivity coefficient are suggested to be chosen as coolants.

5.2. Performance evaluation

In this experiment, the electricity meter 1 is used to monitor the power consumption of the coolant circulation pump and the heating device, and the electricity meter 2 is adopted to monitor the power consumption of other electrical devices, including the cooling water

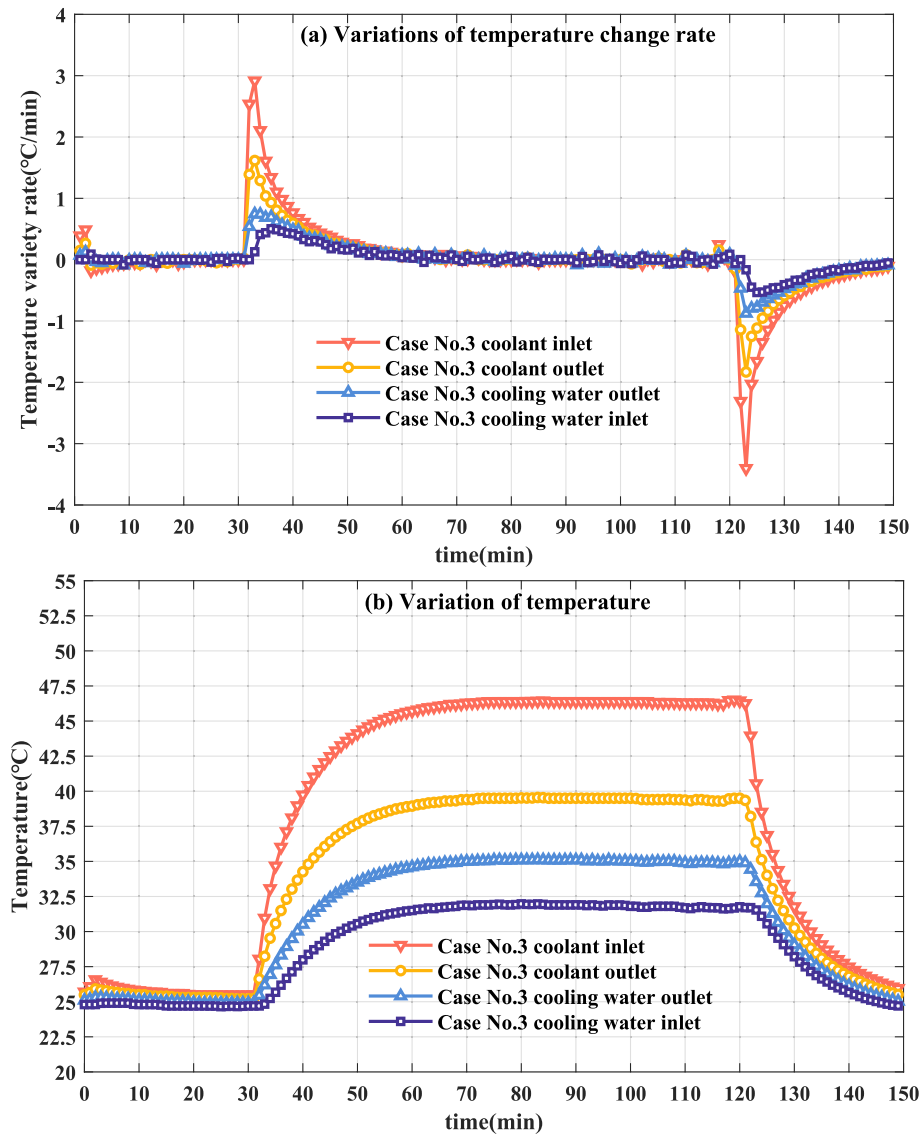


Fig. 6. Fluid temperature variations of Case No.3.

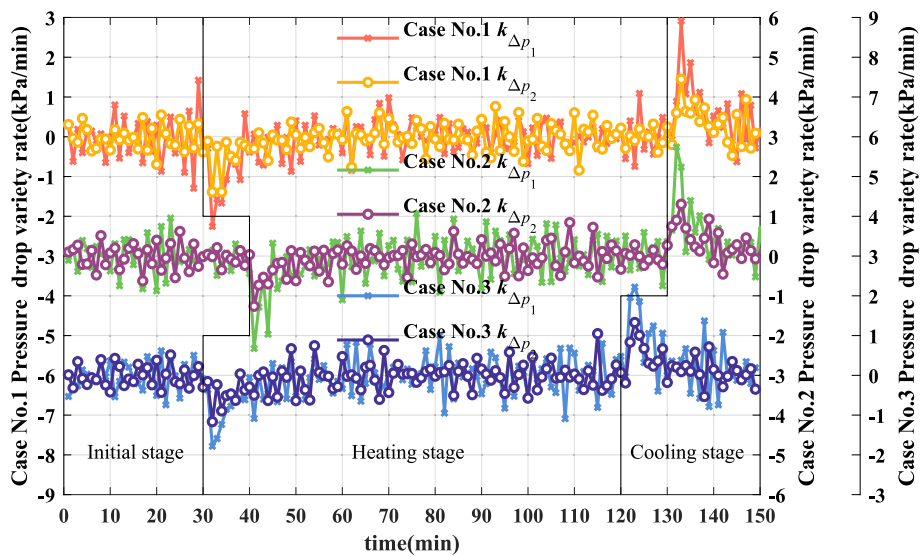


Fig. 7. Variations of change rate of pressure drop with time.

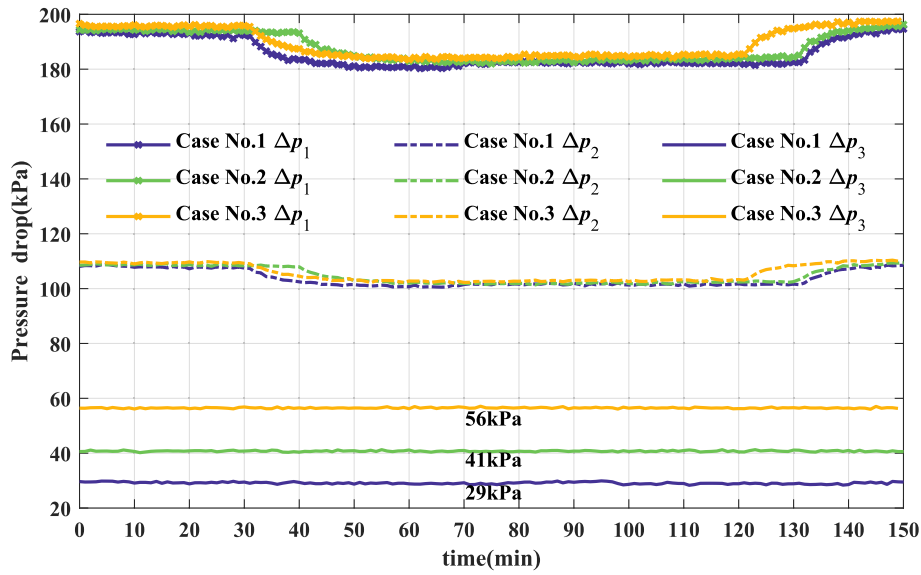


Fig. 8. Variations of pressure drop with time.

Table 3

Inlet and outlet temperatures of the coolant and cooling water.

Case No.	$q_{v1}(\text{m}^3/\text{h})$	$q_{v2}(\text{m}^3/\text{h})$	$T_{c.in} (^{\circ}\text{C})$	$T_{c.out} (^{\circ}\text{C})$	$T_{w.out} (^{\circ}\text{C})$	$T_{w.in} (^{\circ}\text{C})$
1	6.2	4.4	49.9	43.2	39.1	34.7
2	6.2	5.4	48.3	41.5	37.3	33.6
3	6.2	6.4	46.3	39.4	34.9	31.7

Table 4

Temperature difference and logarithmic average temperature difference.

Case No.	$\Delta T_c = T_{c.in} - T_{c.out}$	$\Delta T_w = T_{w.out} - T_{w.in}$	ΔT
1	6.7 $^{\circ}\text{C}$	4.4 $^{\circ}\text{C}$	9.60 $^{\circ}\text{C}$
2	6.8 $^{\circ}\text{C}$	3.7 $^{\circ}\text{C}$	9.36 $^{\circ}\text{C}$
3	6.9 $^{\circ}\text{C}$	3.2 $^{\circ}\text{C}$	9.43 $^{\circ}\text{C}$

circulation pump and cooling tower fan. The detailed power consumptions of different devices in three experiment cases are listed in Tables 7-9.

Here, the cPUE and COP are adopted to evaluate the operation performance of the present SPIC unit.

$$\text{cPUE} = \frac{P_{\text{total}}}{P_{\text{IT}}} \quad (15)$$

Table 5

Heat exchange capacity, heat transfer coefficient, and thermal diffusivity coefficient of three cases.

Case No.	Heat exchange capacity(kW)	Heat transfer coefficient(W/($\text{m}^2\cdot\text{K}$))	Thermal diffusivity coefficient($\times 10^{-8} \text{ m}^2/\text{s}$)
1	23.53	1648	3.05
2	24.08	1729	3.03
3	24.55	1751	3.02

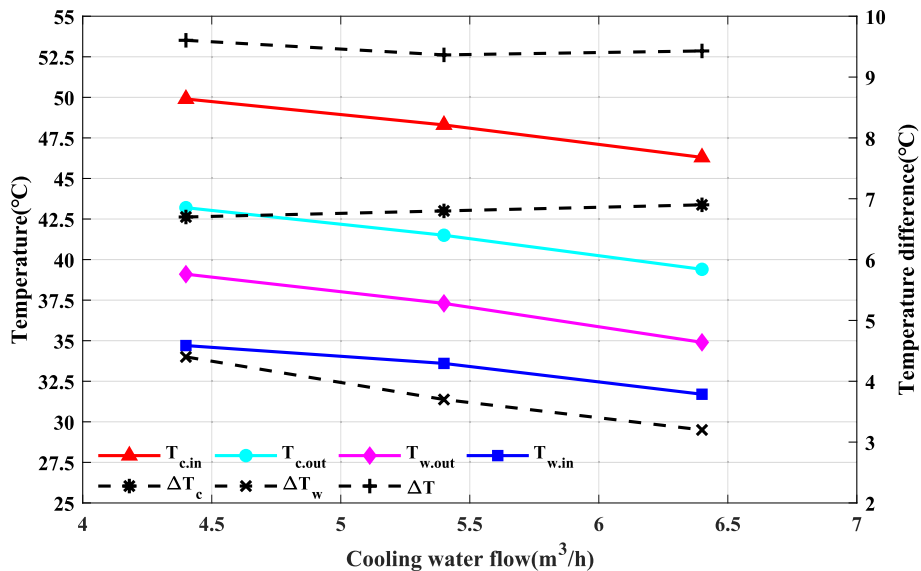


Fig. 9. Temperature variation with cooling water flow rate.

Table 6

Uncertainty of heat exchange capacity, heat transfer coefficient, and thermal diffusivity coefficient of three cases.

Case No.	Heat exchange capacity uncertainty(kW)	Heat transfer coefficient uncertainty(W/(m ² ·K))	Thermal diffusivity coefficient uncertainty (×10 ⁻⁸ m ² /s)
1	1.36	125	0.388
2	1.43	133	0.380
3	1.50	134	0.374

$$\text{COP} = \frac{P_R}{W_R} \quad (16)$$

where, P_{total} is the total power consumption of system, P_{IT} is the power consumption of IT equipment, P_R is the refrigeration capacity of system, and W_R is the power consumption for system refrigeration.

Based on Tables 7–9, it can be seen that the cPUE of the whole SPIC system is around 1.08–1.09. As the PUE value of the traditional air-cooling unit is usually higher than 1.5, it is no doubt that the SPIC unit has a prominent improvement on PUE. With the increase of the cooling water flow rate, the COP gradually decreases from 7.21 to 5.69. This indicates that simply increasing the cooling water flow rate cannot

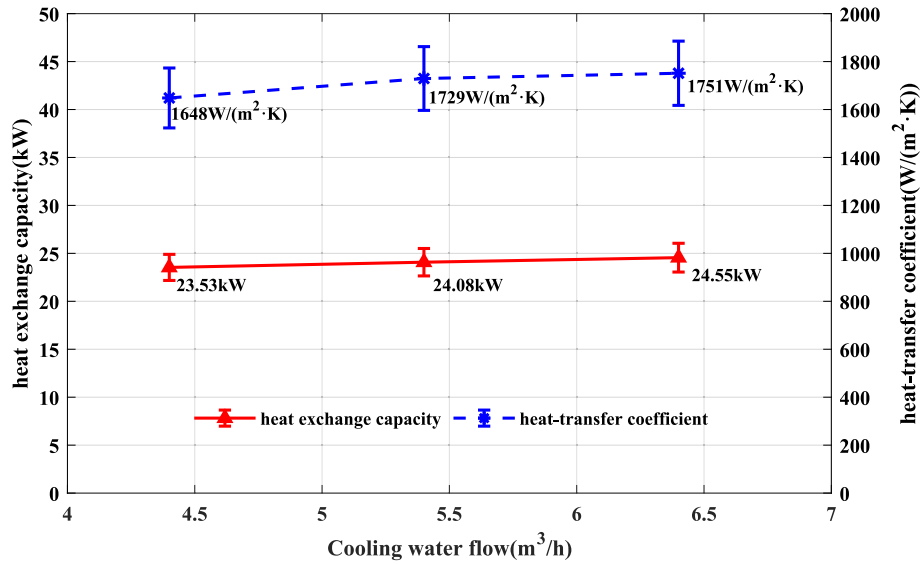


Fig. 10. variations of the heat exchange capacity and heat transfer coefficient with cooling water flow rate.

Table 7

Power consumption in Case No.1.

Time (min)	Meter 1 reading (kWh)	Meter 1 power (kW)	Meter 2 reading (kWh)	Meter 2 power (kW)	P_{total} (kW)	P_{IT} (kW)	P_R (kW)	W_R (kW)	cPUE	COP
0	461.4	—	193.9	—	—	—	—	—	—	—
30	462.0	1.20	194.9	2.00	—	—	—	—	—	—
67	476.8	24.0	195.9	1.62	26.0	24.0	22.5	3.12	1.08	7.21
130	501.9	23.9	198.0	2.00	25.9	23.9	22.4	3.5	1.08	6.40
180	503.4	1.80	199.9	2.28	—	—	—	—	—	—

Table 8

Power consumption in Case No.2.

Time (min)	Meter 1 reading (kWh)	Meter 1 power (kW)	Meter 2 reading (kWh)	Meter 2 power (kW)	P_{total} (kW)	P_{IT} (kW)	P_R (kW)	W_R (kW)	cPUE	COP
0	503.6	—	200.1	—	—	—	—	—	—	—
50	504.7	1.32	201.9	2.16	—	—	—	—	—	—
140	539.9	23.5	204.9	2.00	25.6	23.5	22.04	3.46	1.09	6.37
170	540.7	1.60	205.9	2.00	—	—	—	—	—	—

Table 9

Power consumption in Case No.3.

Time (min)	Meter 1 reading (kWh)	Meter 1 power (kW)	Meter 2 reading (kWh)	Meter 2 power (kW)	P_{total} (kW)	P_{IT} (kW)	P_R (kW)	W_R (kW)	cPUE	COP
0	541.0	—	206.2	—	—	—	—	—	—	—
30	541.8	1.60	207.2	2.00	—	—	—	—	—	—
120	577.6	23.9	210.5	2.20	25.9	23.9	22.2	3.9	1.08	5.69
150	578.5	1.80	211.5	2.00	—	—	—	—	—	—

significantly improve the heat transfer of SPIC units, and will cause an increase of the system energy consumption.

Kanbur et al. (2021) conducted research on single-phase and two-phase immersion cooling systems. In their study, the optimal PUE for single-phase and two-phase immersion cooling systems are 1.05 and 1.08, respectively, when the server power is 15.9 kW. When the total heat load is set as 20 kW, the optimal PUE for the present study is 1.08. The optimal PUE of this study is similar to the optimal PUE of the single-phase immersion cooling system studied by Kanbur et al. (2021). Compared to the optimal PUE of the two-phase immersion cooling system, it is about 3 % higher.

Moreover, the effective power of the circulation pump is also assessed by the following formula.

$$P_e = q_{v1} \Delta p \quad (17)$$

where, P_e is the effective power of the coolant circulation pump, kW, and Δp is the pressure difference between the inlet and outlet of the coolant circulation pump, kPa.

The pump efficiency can be obtained by dividing the effective power of the pump by the power consumption of the pump. The detailed values are listed in Table 10. It is clear to find that the pump efficiency under three operating conditions is around 20 %. This is mainly because the coolant circulation pump operates in a constant frequency. If the variable frequency pumps are used, the energy consumption can be further reduced and system PUE will get further improved.

6. Conclusions

In the present study, a SPIC experimental unit was designed and built. Three different operation conditions were measured by changing cooling water flow rate under a heat load of 20 kW. The inlet and outlet fluid temperatures, the pressure drops of circulation pumps, and the power consumption of the SPIC unit were monitored and analyzed. The cPUE, COP and the effective power of the coolant circulation pump were adopted to evaluate the SPIC system. Some valuable findings were obtained.

- (1) As the cooling water flow rate q_{v2} increases, all monitored temperatures decrease and all monitored pressure drops increase. But the magnitude of changes in each monitored quantity varies. To protect electronic components from temperature and pressure fluctuations in SPIC units, when designing the secondary side cooling subsystem, electronic components should be placed close to the coolant outlet and far from the coolant circulation pump.
- (2) Selecting a liquid with a large density variation with temperature as the coolant can reduce the system flow resistance. The temperature difference between the inlet and outlet of the coolant ΔT_c increases with the increase of q_{v2} due to changes in physical parameters of coolant. ΔT_c may cause uneven temperature distribution in the tank. In addition, to prevent damage to electronic components caused by uneven temperature, liquids with higher thermal diffusivity coefficient can be used as coolants.
- (3) According to the experimental results, on the coolant side, the pressure loss of the heat exchanger accounts for about 56 %, and the pressure loss of pipelines and other equipment accounts for about 44 %.
- (4) In the present experiment test, the measured system's cPUE is 1.08–1.09, and the measured system's COP is 5.69–7.21. This indicates that simply increasing the cooling water flow rate cannot significantly improve the heat transfer performance of SPIC units, and will cause an increase of the system energy consumption. If the performance parameters of the pump are reasonably selected, the operation coefficient of the whole system can get greatly improved. For instance, using frequency

Table 10

Measured power and effective power of coolant circulation pump.

Case No.	Measured power(kW)	Effective power(kW)	Efficiency
1	1.50	0.313	20.87 %
2	1.46	0.317	21.71 %
3	1.70	0.319	18.76 %

conversion control technology can obviously reduce the system energy consumption.

Although some important findings are achieved through the experimental test, this work is mainly focused on the thermal behavior of the SPIC units under different cooling water flow rates. In the future work, effects of different factors on the operation performance of SPIC units will be measured. Moreover, if possible, we will conduct the on-site test for the actual immersion liquid cooling server to obtain the first hand data on the operation performance of the SPIC in data center.

CRedit authorship contribution statement

Xinshan Sun: Writing – original draft, Methodology, Conceptualization. **Zhan Liu:** Writing – original draft, Methodology, Conceptualization, Supervision. **Shenrui Ji:** Writing – original draft, Data curation, Formal analysis, Methodology. **Kaifeng Yuan:** Writing – original draft, Formal analysis, Methodology, Project administration.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This work was supported by the Project Funded by China Postdoctoral Science Foundation (2023M743765).

Data availability

Data will be made available on request.

References

- Brill, K.G., 2007. The invisible crisis in the data center: The economic meltdown of Moore's law. white paper. Uptime Institute, pp. 2–5.
- Cheng, C.C., Chang, P.C., Li, H.C., et al., 2020. Design of a single-phase immersion cooling system through experimental and numerical analysis. *Int. J. Heat Mass Transf.* 160, 120203.
- Duan, Z., Zhan, C., Zhang, X., et al., 2012. Indirect evaporative cooling: Past, present and future potentials. *Renew. Sustain. Energy Rev.* 16 (9), 6823–6850.
- Gajjar, K., 2023. 3D Conjugate Heat Transfer for Single Phase Immersion Cooling of CPU. *Case Studies in Thermal Engineering*. 52, 103728.
- Gładyszewska-Fiedoruk, K., Teleszewski, T.J., 2023. Experimental research on the humidity in a passenger car cabin equipped with an air cooling system-development of a simplified model. *Appl. Therm. Eng.* 220, 119783.
- Huang, Y., Ge, J., Chen, Y., et al., 2023. Natural and forced convection heat transfer characteristics of single-phase immersion cooling systems for data centers. *Int. J. Heat Mass Transf.* 207, 124023.
- Isazadeh, A., Ziviani, D., Claridge, D.E., 2023. Thermal management in legacy air-cooled data centers: An overview and perspectives. *Renew. Sustain. Energy Rev.* 187, 113707.
- Jithin, K.V., Rajesh, P.K., 2022. Numerical analysis of single-phase liquid immersion cooling for lithium-ion battery thermal management using different dielectric fluids. *Int. J. Heat Mass Transf.* 188, 122608.
- Kadam, S.T., Kumar, R., 2014. Twenty first century cooling solution: Microchannel heat sinks. *Int. J. Therm. Sci.* 85, 73–92.
- Kanbur, B.B., Wu, C., Fan, S., et al., 2021. System-level experimental investigations of the direct immersion cooling data center units with thermodynamic and thermoeconomic assessments. *Energy* 217, 119373.
- Khattar, M.K., 2014. Free cooling for data center: honorable mention: industrial facilities, new. *ASHRAE J.* 56 (10), 60–66.

- Landauer, R., 1961. Irreversibility and heat generation in the computing process. *IBM J. Res. Dev.* 5 (3), 183–191.
- Lee, Y.J., Singh, P.K., Lee, P.S., 2015. Fluid flow and heat transfer investigations on enhanced microchannel heat sink using oblique fins with parametric study. *Int. J. Heat Mass Transf.* 81, 325–336.
- Li, G., Sun, Z., Wang, Q., et al., 2023. China's green data center development: Policies and carbon reduction technology path. *Environmental Research*. 231, 116248.
- Li, X., Xu, Z., Liu, S., et al., 2023b. Server performance optimization for single-phase immersion cooling data center. *Appl. Therm. Eng.* 224, 120080.
- Liu, W., Mao, H., Cao, Y., et al., 2021. Long-term energy consumption forecasting for data center industry in China. *IEEE* 4392–4397.
- Liu, Z., Sun, X., Ji, S., et al., 2024. A multi-objective optimization of a single-phase immersion cooling system at cabinet level. *Energ. Buildings* 312, 114221.
- Luiten, W., 2021. Single Phase Passive Hydrocarbon Immersion Cooling of High-power ICs. *IEEE* 1–6.
- Masanet, E., Shehabi, A., Lei, N., et al., 2020. Recalibrating global data center energy-use estimates. *Science* 367, 984–986.
- Mohammed, R.K., Xia, Y., Sahan, R.A., et al., 2012. Performance improvements of air-cooled thermal tool with advanced technologies. *IEEE* 354–361.
- Pambudi, N.A., Yusuf, A.M., Sarifudin, A., 2022. The use of single-phase immersion cooling by using two types of dielectric fluid for data center energy savings. *Energy Eng* 119, 275–286.
- Salim, M., Tozer, R., 2010. Data centers' energy auditing and benchmarking-progress update. *ASHRAE Trans.* 116 (1):109-117.
- Shah, J.M., Bhatt, C., Rachamreddy, P., et al., 2019. Computational form factor study of a 3rd generation open compute server for single-phase immersion cooling. *ASME* 8-15.
- Singh, R., Akbarzadeh, A., Mochizuki, M., 2009. Sintered porous heat sink for cooling of high-powered microprocessors for server applications. *Int. J. Heat Mass Transf.* 52 (9–10), 2289–2299.
- Taddeo, P., Romání, J., Summers, J., et al., 2023. Experimental and numerical analysis of the thermal behaviour of a single-phase immersion-cooled data centre. *Appl. Therm. Eng.* 234, 121260.
- Zhou, G., Li, J., Jia, Z., 2019. Power-saving exploration for high-end ultra-slim laptop computers with miniature loop heat pipe cooling module. *Appl. Energy* 239, 859–875.